

Mathematical Modeling and Numerical Simulation Supplement

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1 Q1 (final: v0)

Abstract— A parsimonious ordinary differential equation framework modeling the coupled evolution of neutron star spin period and magnetic inclination angle under vacuum dipole braking, wind enhancement, inclination decay, and accretion-driven spin-up. The model isolates the dynamical signatures of isolated magnetic-dipole evolution versus binary recycling by encoding phase-dependent torque switching directly into the state derivatives, enabling population-level statistical comparison against observed braking indices and characteristic ages.

1.1 Question and Scope

Question. Can the joint distribution of braking index n and characteristic age τ_c statistically distinguish isolated magnetic-dipole evolution from binary recycling populations, given specific torque model constraints?

First executable model: parsimonious ODE integration of single pulsar tracks for isolated magnetic-dipole spin-down with optional inclination decay and wind braking, embedded in a Monte Carlo population synthesis framework. Binary recycling is represented as an accretion-phase switch with an effective spin-up law. The model excludes glitches, magnetospheric state changes, variable moment of inertia, detailed companion ablation, and full binary evolution.

1.2 Model Assumptions and Applicability

- ASM-001: Stellar geometry, electromagnetic constants, and moment of inertia are absorbed into effective braking constants K_{dipole} and K_{acc} . If violated: Derived constants lose direct physical interpretability and may require explicit microphysical coupling.
- ASM-002: Magnetic inclination evolves smoothly via exponential decay without discontinuous glitches or magnetospheric state transitions. If violated: Smooth ODE integration fails to capture abrupt angular momentum transfers, invalidating continuous derivative assumptions.
- ASM-003: Accretion torque is active exclusively during $t < t_{\text{acc}}$ with nonzero $\dot{M}$; isolated dipole torque applies otherwise. If violated: Simultaneous application of opposing torques produces unphysical period trajectories outside the validated regime.

1.3 Symbols and Units

ID	Symbol	Meaning	Unit
S-P	P	Spin period	s
S-alpha	α	Magnetic inclination angle	rad
S-t	t	Integration time	s
S-P0	P_0	Initial spin period	s
S-alpha0	α_0	Initial inclination angle	rad
S-B0	B_0	Surface magnetic field strength	T
S-Kdip	K_{dipole}	Vacuum dipole braking constant	s T^{-2}
S-eta	η	Wind braking efficiency	dimensionless
S-tau	τ_α	Inclination decay timescale	s
S-Mdot	$\dot{M}$	Mass accretion rate	kg s^{-1}
S-Kacc	K_{acc}	Accretion spin-up constant	$\text{s}^2 \text{kg}^{-1}$
S-tacc	t_{acc}	Accretion phase duration	s
S-Tmax	T_{max}	Maximum simulated age	s
S-Pdotmin	$P_{\text{dot,min}}$	Minimum detectable period derivative	dimensionless
S-Nmc	N_{mc}	Monte Carlo sample count	dimensionless

1.4 Mechanistic Equations

$$\frac{dP}{dt} = \begin{cases} -\frac{K_{\text{acc}}\dot{M}}{\max(P, 0.01)} & \text{if } t < t_{\text{acc}} \wedge \dot{M} > 0 \\ +\frac{K_{\text{dipole}}B_0^2 \sin^2(\alpha)(1+\eta)}{\max(P, 0.01)} & \text{otherwise} \end{cases} \quad (1)$$

Q1-EQ-001 where S-P: Spin period; S-t: Integration time; S-Kacc: Accretion spin-up constant; S-Mdot: Mass accretion rate; S-tacc: Accretion phase duration; S-Kdip: Vacuum dipole braking constant; S-B0: Surface magnetic field strength; S-alpha: Magnetic inclination angle; S-eta: Wind braking efficiency.

$$\frac{d\alpha}{dt} = -\frac{\alpha}{\tau_\alpha} \quad (2)$$

Q1-EQ-002 where S-alpha: Magnetic inclination angle; S-t: Integration time; S-tau: Inclination decay timescale.

$$n(t) = \frac{P(t) \cdot \frac{d^2P}{dt^2}}{\left(\frac{dP}{dt}\right)^2} \quad (3)$$

Q1-EQ-003 where S-P: Spin period.

$$\tau_c(t) = \frac{P(t)}{2 \cdot \frac{dP}{dt}} \quad (4)$$

Q1-EQ-004 where S-P: Spin period.

1.5 Initial Conditions, Boundary Conditions, and Constraints

Initial Conditions

- {'state': 'S-P', 'value': 0.5}
- {'state': 'S-alpha', 'value': 0.7853981633974483}

Boundary Conditions

- {'condition': 'Integration terminates when t reaches T_max or $|dP/dt|$ falls below $Pdot_min$ ', 'type': 'TERMINATION'}

Objective and Constraints

- {'objective': 'Generate distinct P - $Pdot$ - n trajectories for isolated and recycled populations to enable statistical discrimination.'}
- {'constraint': 'Sign convention: positive dP/dt denotes isolated spin-down; negative dP/dt denotes accretion spin-up. Torque branches must not activate simultaneously.'}

1.6 Algorithm

Algorithm

Input

- Initial conditions (P_0 , α_0)
- Approved parameter set
- Scenario configuration

Output

- Time series arrays for $P(t)$, $\alpha(t)$
- Derived diagnostics $n(t)$, $\tau_c(t)$
- Population histograms

Steps

- Initialize state vector with P_0 and α_0
- Select scenario parameter overrides
- Evaluate derivative AST at current time t
- Advance state using fixed-step integrator
- Compute $n(t)$ and $\tau_c(t)$ from derivatives
- Check termination criteria against T_max and $Pdot_min$
- Repeat until termination or N_mc tracks completed

1.7 Parameters, Scenarios, and Numerical Settings

Parameters

- {'symbol_id': 'S-P0', 'value': 0.5}
- {'symbol_id': 'S-alpha0', 'value': 0.7853981633974483}
- {'symbol_id': 'S-B0', 'value': 100000000.0}
- {'symbol_id': 'S-Kdip', 'value': 9.765625e-32}
- {'symbol_id': 'S-eta', 'value': 0.1}
- {'symbol_id': 'S-tau', 'value': 315576000000000.0}
- {'symbol_id': 'S-Mdot', 'value': 630000000000000.0}
- {'symbol_id': 'S-Kacc', 'value': 6.3e-31}
- {'symbol_id': 'S-tacc', 'value': 315576000000000.0}
- {'symbol_id': 'S-Tmax', 'value': 315576000000000.0}
- {'symbol_id': 'S-Pdotmin', 'value': 1e-22}
- {'symbol_id': 'S-Nmc', 'value': 5000.0}

Scenarios

- {'scenario_id': 'isolated_dipole', 'parameter_overrides': {'K_acc': 0.0, 'Mdot': 0.0, 't_acc': 0.0}}
- {'scenario_id': 'binary_recycling', 'parameter_overrides': {'K_acc': 5e-31}}

Solver family. Explicit Runge-Kutta 4th Order.

1.8 Parameter Provenance and Applicability

Approved parameter-set identity. 0f0174315925c8e5b143507cd77c961e1becdad36afebaa31e7f6df12d27946c.

- initial_period (P0) = 0.5 s; role BOUNDARY_CONDITION; status APPROVED_USER_INPUT; source: Birth and Evolution of Isolated Radio Pulsars; DOI 10.1086/501516; document Q1-FGK2006; locator: preprint; applicability: required_condition_1=For isolated or recycled neutron stars at the beginning of the simulated spin-down phase, with period measured in seconds and no glitch-induced period jump at the initial epoch.. uncertainty: not reported.
- initial_inclination (alpha0) = 0.7853981633974483 rad; role BOUNDARY_CONDITION; status APPROVED_MODEL_ASSUMPTION; source: MODEL ASSUMPTION (not a measured or literature value); applicability: required_condition_1=For dipole torque models with approximately aligned magnetic and rotational axes represented by an angle in radians; treated as the starting value before inclination decay.. uncertainty: not reported.

- `initial_surface_magnetic_field (B0)` = 100000000.0 T; role `BOUNDARY_CONDITION`; status `APPROVED_USER_INPUT`; source: Binary and Millisecond Pulsars; DOI 10.12942/lrr-2008-8; document Q1-LORIMER2008; locator: `review_article`; applicability: `required_condition_1`=For isolated neutron stars with approximately dipolar surface fields inferred from timing data, expressed in tesla; not intended for magnetar-like or strongly multipolar fields.. uncertainty: not reported.
- `vacuum_dipole_braking_constant (K_dipole)` = $9.765625 \times 10^{-32} \text{ s T}^{-2}$; role `MODEL_ASSUMPTION`; status `APPROVED_MODEL_ASSUMPTION`; source: `MODEL_ASSUMPTION` (not a measured or literature value); applicability: `required_condition_1`=For the chosen effective spin-down law with magnetic field in tesla and period in seconds, assuming fixed stellar geometry and no unresolved internal braking physics.. uncertainty: not reported.
- `wind_braking_efficiency (eta)` = 0.1 dimensionless; role `MODEL_ASSUMPTION`; status `APPROVED_MODEL_ASSUMPTION`; source: `MODEL_ASSUMPTION` (not a measured or literature value); applicability: `required_condition_1`=For pulsars with magnetospheric wind torque included as a multiplicative enhancement of the dipole term; scenario switches may disable the wind contribution.. uncertainty: not reported.
- `inclination_decay_timescale (tau_alpha)` = 31557600000000.0 s; role `MODEL_ASSUMPTION`; status `APPROVED_USER_INPUT`; source: Binary and Millisecond Pulsars; DOI 10.12942/lrr-2008-8; document Q1-LORIMER2008; locator: `review_article`; applicability: `required_condition_1`=For isolated pulsars modeled with smooth inclination decay, expressed in seconds, excluding glitches and abrupt magnetospheric state changes.. uncertainty: not reported.
- `accretion_rate (Mdot)` = 63000000000000.0 kg s^{-1} ; role `SCENARIO_INPUT`; status `APPROVED_MODEL_ASSUMPTION`; source: `MODEL_ASSUMPTION` (not a measured or literature value); applicability: `required_condition_1`=For recycled pulsars in a steady disk-accretion phase, expressed in kilograms per second, ignoring transient outbursts and companion ablation.. uncertainty: not reported.
- `accretion_braking_constant (K_acc)` = $6.3 \times 10^{-31} \text{ s}^2 \text{ kg}^{-1}$; role `SCENARIO_INPUT`; status `APPROVED_MODEL_ASSUMPTION`; source: `MODEL_ASSUMPTION` (not a measured or literature value); applicability: `required_condition_1`=For the effective accretion law using accretion rate in kilograms per second and dimensionless period derivative; active only during the accretion phase.. uncertainty: not reported.
- `accretion_duration (t_acc)` = 31557600000000.0 s; role `SCENARIO_INPUT`; status `APPROVED_MODEL_ASSUMPTION`; source: `MODEL_ASSUMPTION` (not a measured or literature value); applicability: `required_condition_1`=For binary recycling tracks represented by a finite accretion interval followed by isolated spin-down, expressed in seconds.. uncertainty: not reported.
- `maximum_simulated_age (T_max)` = 31557600000000.0 s; role `BOUNDARY_CONDITION`; status `APPROVED_USER_INPUT`; source: Birth and Evolution of Isolated Radio Pulsars; DOI 10.1086/501516; document Q1-FGK2006; locator: `preprint`; applicability: `required_condition_1`=Numerical endpoint for population synthesis tracks; not a physical observable unless tied to survey selection or evolutionary assumptions.. uncertainty: not reported.

- `minimum_detectable_period_derivative` (`Pdot_min`) = 1e-22 dimensionless; role `BOUNDARY_CONDITION`; status `APPROVED_MODEL_ASSUMPTION`; source: `MODEL_ASSUMPTION` (not a measured or literature value); applicability: `required_condition_1`=Survey-dependent detection boundary for timing observations; used as a stopping condition rather than a physical spin-down parameter.. uncertainty: not reported.
- `monte_carlo_sample_count` (`N_mc`) = 5000.0 dimensionless; role `MODEL_ASSUMPTION`; status `APPROVED_MODEL_ASSUMPTION`; source: `MODEL_ASSUMPTION` (not a measured or literature value); applicability: `required_condition_1`=Numerical control for population synthesis; chosen for computational budget and statistical resolution rather than physical measurement.. uncertainty: not reported.

1.9 Numerical Validation and Model-Internal Results

Validation Plan

- {'method': 'Analytical limit check: verify $d\alpha/dt$ converges to zero as t approaches infinity', 'metric': 'Relative error < 1e-6'}
- {'method': 'Scenario isolation test: confirm isolated_dipole yields strictly positive dP/dt throughout', 'metric': 'Sign consistency across all t '}
- {'method': 'Recycling transition test: verify dP/dt switches sign at t_{acc} and returns to positive post-accretion', 'metric': 'Branch activation count matches scenario definition'}
- Execution `sim-ef92f26db7c14196aaab75759811335e`; `NUMERICAL_SIMULATION`; `SIMULATED`; `NOT_EMPIRICAL`; relation `CONSTRAINED`; summary: Within the audited reduced ODE, the isolated track increases from $P=0.500$ s to 0.662 s, whereas the recycling track reaches $P_{min}=0.226$ s during accretion and ends at 0.281 s after resumed dipole spin-down. This establishes model-internal directional separation, but does not by itself validate population-level $n\text{-}\tau_c$ discrimination because the executable MathIR contains two deterministic tracks rather than the documented Monte Carlo outer loop.

1.10 Hypothesis Iteration Lineage

- $v0$: `CONSTRAINED` (Within the audited reduced ODE, the isolated track increases from $P=0.500$ s to 0.662 s, whereas the recycling track reaches $P_{min}=0.226$ s during accretion and ends at 0.281 s after resumed dipole spin-down. This establishes model-internal directional separation, but does not by itself validate population-level $n\text{-}\tau_c$ discrimination because the executable MathIR contains two deterministic tracks rather than the documented Monte Carlo outer loop.)

1.11 Limitations, Non-Empirical Disclosure, and References

All values in this section are model-internal numerical simulations (`NUMERICAL_SIMULATION`; `SIMULATED`; `NOT_EMPIRICAL`), not empirical observations.

Limitations

- Excludes glitch events, magnetospheric state transitions, and variable moment of inertia.

- Assumes fixed stellar geometry and absorbs unresolved electromagnetic factors into effective constants.
- Observational selection effects and survey detection thresholds are not dynamically coupled to the simulation loop.

References

- {'citation': 'Feng, G., et al. (2006). Birth and Evolution of Isolated Radio Pulsars. *ApJ*, 646, 381. DOI: 10.1086/501516', 'source': 'Q1-FGK2006'}
- {'citation': 'Lorimer, D. R. (2008). Binary and Millisecond Pulsars. *Living Rev. Relativity*, 11, 8. DOI: 10.12942/lrr-2008-8', 'source': 'Q1-LORIMER2008'}

2 Q2 (final: v1)

Abstract— A spatially resolved spherical radial thermal model for a massive neutron star core is specified as an executable PDE branch. The model replaces the parent lumped thermal state with a one-dimensional radial heat-transport field in the canonical coordinate x , where x represents the physical radial coordinate r normalized by the stellar radius. Phase-transition latent heat and modified neutrino cooling are represented through parameterized volumetric source terms. Scenario A is standard cooling without latent heat, Scenario B adds localized latent-heat release, and Scenario C modifies the cooling coefficient to emulate superfluid suppression. The declared observable outputs are radial temperature trajectories and a surface flux proxy at $x=1$.

2.1 Question and Scope

Question. How does the radial thermal evolution of a massive neutron star differ when spin-down-associated core compression triggers a localized phase-transition heat source, compared with standard cooling and with a superfluid-modified cooling scenario?

Executable spherical radial thermal PDE branch for controlled numerical simulation. The model covers radial temperature evolution, a parameterized phase-transition heat source, volumetric cooling, center regularity, and a fixed scaled surface temperature boundary. It does not solve spin dynamics, density evolution, full microphysics, atmosphere radiative transfer, or observational inference.

2.2 Model Assumptions and Applicability

- Q2-ASM-001: The star is spherically symmetric and the executable PDE uses the canonical coordinate x as the physical radial coordinate r normalized by the stellar radius. If violated: Non-spherical structure or multidimensional transport would not be represented, biasing radial temperature and surface-flux estimates.
- Q2-ASM-002: Spin-down induced core compression and phase-transition onset are represented by a prescribed localized latent-heat source rather than dynamically solved spin, density, or moment-of-inertia equations. If violated: Feedback between rotation, compression, and transition timing is omitted, so scenario attribution becomes phenomenological.

- Q2-ASM-003: Thermal microphysics is reduced to scaled constant transport, heat-capacity, and source coefficients. If violated: Temperature- and density-dependent conductivity, heat capacity, and neutrino emissivity would change cooling rates and scenario separation.
- Q2-ASM-004: The surface boundary is approximated by a fixed scaled surface temperature representing atmosphere and radiative-transfer effects. If violated: An inaccurate surface boundary would alter the surface gradient, flux proxy, and inferred luminosity.
- Q2-ASM-005: Units are normalized so that the temperature field is scaled by a reference core temperature, $x=1$ corresponds to the stellar radius, and $t=1$ corresponds to a characteristic cooling time. If violated: Direct physical-unit comparison requires rescaling; mismatched scaling would distort comparison with cooling curves.

2.3 PDE Model Class

System type. SPHERICAL_RADIAL_THERMAL.

Trusted adapter. spherical_radial_thermal.

2.4 Spatial Domain and Field Variables

Domain and mesh

- Domain: $x=[0.0, 1.0]$
- Grid: $nx=33$

Fields

- T (T); unit 10^9 K

2.5 PDE Initial and Boundary Conditions

Initial condition

- Sampled values on the normalized mesh.
- Time span: 0.0; 10.0

Boundary condition types

- left: SPHERICAL_ORIGIN_REGULARITY
- right: DIRICHLET

2.6 Discretization and Stability

Discretization. FINITE_DIFFERENCE_SPHERICAL_RADIAL.

Time integration. EXPLICIT_EULER; time step 0.005.

2.7 Numerical Verification

The fixed PDE adapter reports finite-field checks and the family-specific stability or linear-residual diagnostic. A result is qualified only when the declared checks pass; all outcomes remain model-internal simulations.

2.8 Symbols and Units

ID	Symbol	Meaning	Unit
T	T	Radial temperature field scaled by a reference core temperature	$10^9 K$
x	x	Canonical radial coordinate representing physical radius r normalized by stellar radius	RNS
t	t	Time scaled by a characteristic cooling time	t char
source	S	Volumetric heat source and sink term in scaled temperature per time	$10^9 K/t_{char}$
kappa	κ	Effective radial thermal diffusion coefficient in scaled units	$R_N S^2/t_{char}$
c_v	c_v	Volumetric heat capacity in scaled units	energy/(R_NS ³ *10 ⁹ K)
latent_amp	L_{amp}	Amplitude of latent-heat release from the phase-transition layer	$10^9 K/t_{char}$
cooling_amp	ϵ_{amp}	Amplitude of volumetric neutrino cooling coefficient	$1/t_{char}$
transition_center	x_{tr}	Normalized radial center of the phase-transition layer	RNS
transition_width	σ_{tr}	Normalized radial width of the phase-transition layer	RNS
decay_rate	λ	Inverse time scale for transition-source decay	$1/t_{char}$
surface_temperature	T_s	Scaled surface temperature imposed at x=1	$10^9 K$

2.9 Mechanistic Equations

$$c_v \frac{\partial T}{\partial t} = \frac{1}{x^2} \frac{\partial}{\partial x} \left(x^2 \kappa \frac{\partial T}{\partial x} \right) + S \quad (5)$$

Q2-EQ-001 where c_v: Volumetric heat capacity in scaled units; T: Radial temperature field scaled by a reference core temperature; t: Time scaled by a characteristic cooling time; x: Canonical radial coordinate representing physical radius r normalized by stellar radius; kappa: Effective radial thermal diffusion coefficient in scaled units; source: Volumetric heat source and sink term in scaled temperature per time.

$$S = L_{amp} \exp \left[-\frac{(x - x_{tr})^2}{2\sigma_{tr}^2} \right] \exp(-\lambda t) - \epsilon_{amp} T \quad (6)$$

Q2-EQ-002 where source: Volumetric heat source and sink term in scaled temperature per time; latent_amp: Amplitude of latent-heat release from the phase-transition layer; x: Canonical radial coordinate representing physical radius r normalized by stellar radius; transition_center: Normalized radial center of the phase-transition layer; transition_width: Normalized radial width of the phase-transition layer; decay_rate: Inverse time scale for transition-source decay; t: Time scaled by a characteristic cooling time; cooling_amp: Amplitude of volumetric neutrino cooling coefficient; T: Radial temperature field scaled by a reference core temperature.

$$\left. \frac{\partial T}{\partial x} \right|_{x=0} = 0, T(1, t) = T_s \quad (7)$$

Q2-EQ-003 where T: Radial temperature field scaled by a reference core temperature; x: Canonical radial coordinate representing physical radius r normalized by stellar radius; t: Time scaled by a characteristic cooling time; surface_temperature: Scaled surface temperature imposed at x=1.

2.10 Initial Conditions, Boundary Conditions, and Constraints

Initial Conditions

- {'condition_id': 'Q2-IC-001', 'field_id': 'T', 'profile': 'UNIFORM', 'description': 'Uniform scaled initial core temperature across the radial domain.'}

Boundary Conditions

- {'boundary_id': 'Q2-BC-001', 'side': 'left', 'type': 'SPHERICAL_ORIGIN_REGULARITY', 'description': 'Regularity at the spherical origin $x=0$.'}
- {'boundary_id': 'Q2-BC-002', 'side': 'right', 'type': 'DIRICHLET', 'value_parameter': 'surface_temperature', 'description': 'Fixed scaled surface temperature at $x=1$.'}

Objective and Constraints

- {'objective_id': 'Q2-OBJ-001', 'statement': 'Compare radial temperature trajectories and the surface flux proxy among standard, phase-transition, and superfluid-modified scenarios.'}
- {'objective_id': 'Q2-OBJ-002', 'statement': 'Determine whether localized latent-heat release produces a distinguishable delay or deviation in cooling relative to standard cooling.'}
- {'constraint_id': 'Q2-CON-001', 'statement': 'Use only the registered spherical radial thermal PDE adapter with a uniform grid and explicit time integration.'}
- {'constraint_id': 'Q2-CON-002', 'statement': 'Maintain non-negative diffusion coefficient, positive heat capacity, and finite field values throughout the declared time span.'}

2.11 Algorithm

Algorithm

Input

- scenario parameter values
- spatial_domain
- grid
- initial_condition
- boundary_conditions

Output

- radial temperature field $T(x,t)$
- surface flux proxy at $x=1$
- scenario comparison diagnostics

Steps

- Select scenario parameter values.
- Initialize the uniform scaled temperature field.
- Apply spherical origin regularity and the surface Dirichlet condition.
- Advance the spherical radial heat equation with the registered explicit integrator.
- Evaluate stability and finiteness checks.
- Extract radial profiles and the surface flux proxy.
- Compare scenario tracks against the declared falsification criteria.

2.12 Parameters, Scenarios, and Numerical Settings

Parameters

- {'parameter_id': 'Q2-PRM-001', 'symbol_id': 'kappa', 'base_value': 0.01, 'evidence_status': 'model_assumption', 'source': 'Scaled effective radial thermal diffusion chosen for explicit spherical stability.'}
- {'parameter_id': 'Q2-PRM-002', 'symbol_id': 'c_v', 'base_value': 1.0, 'evidence_status': 'model_assumption', 'source': 'Scaled volumetric heat capacity reference.'}
- {'parameter_id': 'Q2-PRM-003', 'symbol_id': 'latent_amp', 'base_value': 0.0, 'evidence_status': 'scenario_input', 'source': 'No latent heat in the standard scenario; positive in phase-transition scenarios.'}
- {'parameter_id': 'Q2-PRM-004', 'symbol_id': 'cooling_amp', 'base_value': 0.02, 'evidence_status': 'model_assumption', 'source': 'Scaled neutrino cooling coefficient.'}
- {'parameter_id': 'Q2-PRM-005', 'symbol_id': 'transition_center', 'base_value': 0.35, 'evidence_status': 'model_assumption', 'source': 'Normalized core radius where the phase transition is concentrated.'}
- {'parameter_id': 'Q2-PRM-006', 'symbol_id': 'transition_width', 'base_value': 0.12, 'evidence_status': 'model_assumption', 'source': 'Normalized radial width of the transition layer.'}
- {'parameter_id': 'Q2-PRM-007', 'symbol_id': 'decay_rate', 'base_value': 0.6, 'evidence_status': 'model_assumption', 'source': 'Inverse time scale for latent-heat release decay.'}
- {'parameter_id': 'Q2-PRM-008', 'symbol_id': 'surface_temperature', 'base_value': 0.1, 'evidence_status': 'model_assumption', 'source': 'Scaled surface temperature representing the atmosphere boundary.'}

Scenarios

- {'scenario_id': 'Q2-SCN-A', 'name': 'Standard cooling', 'description': 'No phase-transition latent heat and standard scaled neutrino cooling.', 'parameter_overrides': {'latent_amp': 0.0, 'cooling_amp': 0.02, 'surface_temperature': 0.1}}
- {'scenario_id': 'Q2-SCN-B', 'name': 'Phase transition cooling', 'description': 'Localized latent-heat release associated with spin-down induced core compression.', 'parameter_overrides': {'latent_amp': 0.3, 'cooling_amp': 0.02, 'surface_temperature': 0.1}}
- {'scenario_id': 'Q2-SCN-C', 'name': 'Phase transition with superfluidity', 'description': 'Latent-heat release with suppressed neutrino cooling representing superfluid modification.', 'parameter_overrides': {'latent_amp': 0.3, 'cooling_amp': 0.01, 'surface_temperature': 0.1}}

Solver family. spherical_radial_thermal.

2.13 Parameter Provenance and Applicability

Parameter status. This legacy model used inline assumptions; it has no approved external parameter set.

2.14 Numerical Validation and Model-Internal Results

Validation Plan

- {'check_id': 'Q2-VAL-001', 'description': 'Preflight stability: the explicit spherical diffusion number remains below the registered limit.', 'status': 'required_before_execution'}
- {'check_id': 'Q2-VAL-002', 'description': 'Scenario separation: phase-transition scenarios produce different radial temperature histories and surface flux proxy values from the standard scenario.', 'status': 'required_after_execution'}
- {'check_id': 'Q2-VAL-003', 'description': 'Boundary consistency: center regularity and surface Dirichlet condition are satisfied without non-finite values.', 'status': 'required_after_execution'}
- {'check_id': 'Q2-VAL-004', 'description': 'Observational relevance: the surface flux proxy is mapped to luminosity only after external scaling and calibration.', 'status': 'postprocessing'}
- Execution sim-1b3a94f9eebb4426b07d75c423ce2def; NUMERICAL_SIMULATION; SIMULATED; NOT_EMPIRICAL; relation INCONCLUSIVE; summary: Q2 v1 baseline spherical-radial thermal PDE completed with finite output, explicit stability, spherical-origin regularity, and boundary checks passing; this baseline-only run does not establish scenario separation or empirical pulsar validation.

2.15 Hypothesis Iteration Lineage

- v0: SUPPORTED_WITHIN_MODEL (The audited reduced ODE crosses $\rho_{\text{crit}}=5.12\text{e}17 \text{ kg/m}^3$ at approximately $9.85\text{e}11 \text{ s}$ in all spin-down cases. At $3\tau_{\text{cool}}$, the standard, phase-transition, and phase-transition-plus-superfluid scenarios yield surface temperatures of about $4.98\text{e}4 \text{ K}$, $5.32\text{e}4 \text{ K}$, and $2.36\text{e}5 \text{ K}$, respectively; phase_fraction

reaches 1 only in the two transition scenarios. The mechanism therefore produces distinguishable model-internal thermal tracks, conditional on the effective cooling, latent-heat, and compression proxies.)

- v1: INCONCLUSIVE (Q2 v1 baseline spherical-radial thermal PDE completed with finite output, explicit stability, spherical-origin regularity, and boundary checks passing; this baseline-only run does not establish scenario separation or empirical pulsar validation.)

2.16 Limitations, Non-Empirical Disclosure, and References

All values in this section are model-internal numerical simulations (NUMERICAL_SIMULATION; SIMULATED; NOT_EMPIRICAL), not empirical observations.

Limitations

- The model does not solve spin evolution, moment of inertia, or central density dynamically; spin-down compression is parameterized by a prescribed source.
- Microphysical equation-of-state effects, latent heat, neutrino emissivity, and superfluid suppression are represented by scaled coefficients rather than density-temperature tables.
- The surface boundary is simplified; atmosphere radiative transfer and gravitational redshift are not solved.
- Spherical symmetry and a uniform radial grid exclude multidimensional effects and sharp phase-boundary dynamics.
- No observational data are ingested; comparison to X-ray cooling curves requires external calibration and uncertainty treatment.

References

- Standard neutron-star cooling theory and cooling-curve literature for model comparison.
- X-ray observations of isolated neutron stars and massive pulsars from Chandra and XMM-Newton archives as external comparator context.
- Equation-of-state and phase-transition literature for neutron-star cores as parameter motivation.